

Bivariate Distributions

- a random experiment whose outcome is a scalar $\rightarrow$ **univariate** RV
- two random experiments jointly each of whose outcome is a scalar,
or a random experiment whose outcome is a pair of two scalars $\rightarrow$ **bivariate** RV

Let (X, Y) be a bivariate RV. Then

- the range: $S \subseteq \mathbb{R}^2$
- the range of X : $S_X \subseteq \mathbb{R}$, and the range of Y : $S_Y \subseteq \mathbb{R}$
- it holds that $S \subseteq S_X \times S_Y$

Remark. X and Y can be dependent. Therefore, it may hold that $S \subset S_X \times S_Y$.

1 Bivariate Distributions of Discrete Type

1.1 Definitions

Definition 1.1 (Discrete Bivariate Random Variable).

A bivariate random variable is said to be of the **discrete type** if its range is *finite* or *countably infinite*.

Definition 1.2 (Joint Probability Mass Function).

A function $f: S \subset \mathbb{R}^2 \rightarrow [0, 1]$ is called the **joint probability mass function** (joint pmf) of a DRV (X, Y) with range S if it satisfies:

- $f(x, y) > 0$ for $(x, y) \in S$, and $f(x, y) = 0$ for $(x, y) \notin S$
- $\sum_{(x, y) \in S} f(x, y) = 1$
- $\mathbb{P}[(X, Y) \in A] = \sum_{(x, y) \in A} f(x, y)$, where $A \subseteq S$

Then S is called the **support** or **space** of X .

Remark. From the third property, we have:

$$\mathbb{P}(X = x, Y = y) = f(x, y)$$

Remark. The summation of the pmf over a set $A \subseteq S$ can be split into two single summations.

Define $A_X := \{x \mid (x, y) \in A\}$, and $A_Y(x) := \{y \mid (x, y) \in A\}$ for $x \in A_X$.

$$\mathbb{P}[(X, Y) \in A] = \sum_{(x, y) \in A} f(x, y) = \sum_{x \in A_X} \sum_{y \in A_Y(x)} f(x, y)$$

Definition 1.3 (Marginal Probability Mass Function).

Let (X, Y) has a joint pmf $f(x, y)$ with space S . The pmf of X alone, called the **marginal probability mass function** (marginal pmf) of X , is defined as:

$$f_X(x) := \mathbb{P}(X = x) = \sum_{y \in S_Y(x)} f(x, y)$$

where

$$S_Y(x) := \{y \mid (x, y) \in S\} \quad \text{for } x \in S_X$$

1.2 Independence

Definition 1.4 (Independence).

$X \in S_X$ and $Y \in S_Y$ are said to be **independent** iff

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$$

or equivalently

$$f(x, y) = f_X(x)f_Y(y)$$

When X and Y are independent, we have:

$$S = S_X \times S_Y \quad \text{is rectangular}$$

$$S_X = S_X(y) \quad \text{for } \forall y \in S_Y$$

$$S_Y = S_Y(x) \quad \text{for } \forall x \in S_X$$

However, the converse is not true.

1.3 Mathematical Expectations

Definition 1.5 (Mathematical Expectation).

Let X and Y be discrete RVs with joint pmf $f: S \rightarrow [0, 1]$. The **mathematical expectation** of a function $g(X, Y)$ is

$$\mathbb{E}[g(X, Y)] := \sum_{(x, y) \in S} g(x, y)f(x, y)$$

Theorem 1.1.

Two ways to compute the mean of X :

$$\mathbb{E}[X] = \sum_{x \in S_X} xf_X(x) = \sum_{(x, y) \in S} xf(x, y)$$

2 The Corelation Coefficient

Definition 2.1 (Covariance).

The **covariance** of two discrete RVs X and Y is defined as

$$\sigma_{XY} = \text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

- $\text{Cov}(X, Y) < 0 \implies X$ and Y are **negatively correlated**
- $\text{Cov}(X, Y) > 0 \implies X$ and Y are **positively correlated**
- $\text{Cov}(X, Y) = 0 \implies X$ and Y are **uncorrelated**

$$\mathbb{E}[XY] = \mathbb{E}[X \cdot \mathbb{E}[Y | X]] = \mathbb{E}[Y \cdot \mathbb{E}[X | Y]]$$

Meaning: $\mathbb{E}[XY] = \mathbb{E}[g(X)]$, where $g(x) = x \cdot \mathbb{E}[Y | X = x]$.

Proof.

$$\mathbb{E}[XY] = \sum_{(x, y) \in S} xyf(x, y) = \sum_{x \in S_X} x \left(\sum_{y \in S_Y(x)} y \cdot f_{Y|X}(y | x) \right) f(x) = \sum_{x \in S_X} x \mathbb{E}[Y | X = x] f(x) = \mathbb{E}[X \cdot \mathbb{E}[Y | X = X]]$$

□

Theorem 2.1 (Properties of Covariance).

Let X, Y, Z be RVs, and let a, b, c, d be constants. Then:

- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$
- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- $\text{Cov}(X, a) = 0$

Theorem 2.2.

In terms of two RVs X and Y , generally,

$$\begin{aligned} \text{Independent} &\implies \text{Uncorrelated} && \text{i.e. } \text{Cov}(X, Y) = 0 \\ \text{Independent} &\not\Leftarrow \text{Uncorrelated} \end{aligned}$$

Proof (Independence implies uncorrelation).

$$f(x, y) = f_X(x)f_Y(y) \implies \mathbb{E}[XY] = \sum_{x \in S_X} \sum_{y \in S_Y} xy f_X(x) f_Y(y) = \mathbb{E}[X]\mathbb{E}[Y] \implies \text{Cov}(X, Y) = 0$$

□

Definition 2.2 (Correlation Coefficient).

The **correlation coefficient** of two discrete RVs X and Y is defined as

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \in [-1, 1]$$

provided that $\sigma_X \sigma_Y \neq 0$.

Remark. ρ is a normalized version of $\text{Cov}(X, Y)$.

Theorem 2.3.

ρ provides a normalized measure of the strength of the linear relationship between X and Y .

$$|\rho| = 1 \iff \exists c \text{ s.t. } Y - \mathbb{E}[Y] = c(X - \mathbb{E}[X])$$

where c shares the same sign as ρ and

$$|c| = \sqrt{\frac{\text{Var}(Y)}{\text{Var}(X)}}$$

Theorem 2.4 (Cauchy-Schwarz Inequality).

For any two RVs X and Y ,

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$$

Proof (Cauchy-Schwarz Inequality).

Let $U = X - \mathbb{E}[X]$ and $V = Y - \mathbb{E}[Y]$. Let $f(t) = \mathbb{E}[(U + tV)^2]$. Then

$$f(t) = \mathbb{E}[U^2] + 2t\mathbb{E}[UV] + t^2\mathbb{E}[V^2] = \text{Var}(X) + 2t\text{Cov}(X, Y) + t^2\text{Var}(Y) \geq 0$$

The discriminant of $f(t)$ is

$$\Delta = 4\text{Cov}(X, Y)^2 - 4\text{Var}(X)\text{Var}(Y) = 4\text{Cov}(X, Y)^2 - 4\text{Var}(X)\text{Var}(Y) \leq 0$$

□

3 Conditional Distributions

Definition 3.1 (Conditional Probability Mass Function).

The **conditional pmf** of X given $Y = y$ is defined as

$$f_{X|Y}(x | y) := \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{for } x \in S_X(y)$$

provided that $f_Y(y) > 0$.

Proof.

$$f_{X|Y}(x | y) = \mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

□

Remark. Easy to check that conditional pmf is well-defined.

Remark. When X and Y are independent, we have:

$$f_{X|Y}(x | y) = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x)$$

Definition 3.2 (Conditional Mathematical Expectation).

Let $g(Y)$ be a function of Y . The **conditional expectation** of $g(Y)$ given $X = x$ is defined as

$$\mathbb{E}[g(Y) | X = x] := \sum_{y \in S_Y(x)} g(y) \cdot f_{Y|X}(y | x)$$

where $x \in S_X$.

- Conditional mean: $\mathbb{E}[Y | X = x]$
- Conditional variance: $\text{Var}(Y | X = x) := \mathbb{E}[(Y - \mathbb{E}[Y | X = x])^2 | X = x]$

Remark.

$$\text{Var}(Y | X = x) = \mathbb{E}[Y^2 | X = x] - (\mathbb{E}[Y | X = x])^2$$

Theorem 3.1.

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y = Y]] = \sum_{y \in S_Y} \mathbb{E}[X | Y = y] f_Y(y) = \sum_{(x,y) \in S} x f_{X,Y}(x, y)$$

4 Examples

4.1 Trinomial Distribution

Description A random experiment that has three *mutually exclusive* and *exhaustive* outcomes is repeated n independent times, and the probabilities of each outcome remain the same for each repetition. Such n repetitions can be called a trinomial experiment. Let X , Y , and $n - X - Y$ be the number of the three outcomes, respectively.

Definition 4.1 (Trinomial Distribution).

Parameters $n = 1, 2, 3, \dots$ $p_X \in [0, 1]$ $p_Y \in [0, 1]$ s.t. $p_X + p_Y \leq 1$.

(X, Y) is said to have a **trinomial distribution**, if the **joint pmf** is given by

$$f_{X,Y}(x, y) = \binom{n}{x, y, n-x-y} p_X^x p_Y^y (1 - p_X - p_Y)^{n-x-y} = \frac{n!}{x!y!(n-x-y)!} p_X^x p_Y^y (1 - p_X - p_Y)^{n-x-y}$$

The **marginal pmf** of X is given by

$$f_X(x) = \binom{n}{x} p_X^x (1 - p_X)^{n-x}$$

Namely, $X \sim B(n, p_X)$, $Y \sim B(n, p_Y)$.

Remark. Trinomial Expansion

$$(a + b + c)^n = \sum_{x=0}^n \sum_{y=0}^{n-x} \binom{n}{x, y, n-x-y} a^x b^y c^{n-x-y} = \sum_{x=0}^n \sum_{y=0}^{n-x} \frac{n!}{x!y!(n-x-y)!} a^x b^y c^{n-x-y}$$

Theorem 4.1 (Trinomial Distribution).

The conditional pmf of Y given $X = x$ is

$$f_{Y|X}(y | x) = \binom{n-x}{y} \left(\frac{p_Y}{1-p_X} \right)^y \left(1 - \frac{p_Y}{1-p_X} \right)^{n-x-y}$$

Namely, $(Y | X = x) \sim B\left(n-x, \frac{p_Y}{1-p_X}\right)$, and $(X | Y = y) \sim B\left(n-y, \frac{p_X}{1-p_Y}\right)$.

The covariance of X and Y can be computed as

$$\mathbb{E}[XY] = n(n-1) \cdot p_X p_Y \quad \text{Cov}(X, Y) = -n \cdot p_X p_Y$$

Proof (Covariance of trinomial distribution).

Method 1 Conditional Expectation Method

$$\begin{aligned} \mathbb{E}[Y^2] &= (n^2 - n) \cdot p_Y^2 + n \cdot p_Y \\ \mathbb{E}[XY] &= \mathbb{E}[Y \cdot \mathbb{E}[X | Y]] = \frac{p_X}{1-p_Y} \mathbb{E}[Y(n-Y)] = n(n-1) \cdot p_X p_Y \end{aligned}$$

Method 2 Indicator Variable Method

For each trial i , define the indicator variables X_i and Y_i :

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th trial is the outcome of } X \\ 0 & \text{otherwise} \end{cases} \quad Y_i = \begin{cases} 1 & \text{if the } i\text{-th trial is the outcome of } Y \\ 0 & \text{otherwise} \end{cases}$$

Then X_i and Y_j are independent for $i \neq j$, and $X_i Y_i = 0$. Therefore,

$$\mathbb{E}[XY] = \mathbb{E}\left[\sum_{i=1}^n X_i \sum_{i=1}^n Y_i\right] = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[X_i Y_j] = \sum_{i \neq j} \mathbb{E}[X_i] \mathbb{E}[Y_j] = \sum_{i \neq j} p_X p_Y = n(n-1) p_X p_Y$$

□

5 Bivariate Distributions of Continuous Type

Definition 5.1 (Joint Probability Density Function).

A function $f: S \subseteq \mathbb{R}^2 \rightarrow [0, +\infty)$ is called the **joint probability density function** (joint pdf) of a CRV (X, Y) with range S if it satisfies:

- $f(x, y) > 0$ for $(x, y) \in S$, and $f(x, y) = 0$ for $(x, y) \notin S$
- $\iint_S f(x, y) \, dx \, dy = 1$
- $\mathbb{P}[(X, Y) \in A] = \iint_A f(x, y) \, dx \, dy$, where $A \subseteq S$

Then S is called the **support** or **space** of X .

Definition 5.2 (Marginal Probability density Function).

Let (X, Y) has a joint pdf $f(x, y)$ with space S . The pdf of X alone, called the **marginal probability density function** (marginal pdf) of X , is defined as:

$$f_X(x) := \int_{S_Y(x)} f(x, y) \, dy \quad \text{for } x \in S_X$$

where

$$S_Y(x) := \{y \mid (x, y) \in S\} \quad \text{for } x \in S_X$$

5.1 Mathematical Expectations

Definition 5.3 (Mathematical Expectation).

Let X and Y be CRVs with joint pdf $f: S \rightarrow [0, +\infty)$. The **mathematical expectation** of a function $g(X, Y)$ is

$$\mathbb{E}[g(X, Y)] := \iint_S g(x, y) f(x, y) \, dx \, dy$$

Theorem 5.1.

$$\mathbb{E}[X] = \int_{S_X} x f_X(x) \, dx = \iint_S x f(x, y) \, dx \, dy = \int_{S_Y} \mathbb{E}[X \mid Y = y] f_Y(y) \, dy = \mathbb{E}[\mathbb{E}[X \mid Y = Y]]$$

5.2 Covariance & Correlation Coefficient

Definition 5.4 (Independence).

$X \in S_X$ and $Y \in S_Y$ are said to be **independent** iff

$$f(x, y) = f_X(x) f_Y(y)$$

If X and Y are independent, then

$$S = S_X \times S_Y$$

Definition 5.5 (Covariance & Correlation Coefficient).

Let X and Y be CRVs.

$$\begin{aligned}\text{Cov}(X, Y) &:= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \quad \text{for } \text{Var}(X)\text{Var}(Y) \neq 0$$

5.3 Conditional Distributions

Definition 5.6 (Conditional Probability Density Function).

The **conditional pdf** of X given $Y = y$ is defined as

$$f_{X|Y}(x | y) := \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{for } x \in S_X(y)$$

provided that $f_Y(y) > 0$.

Definition 5.7 (Conditional Mathematical Expectation).

Let $g(Y)$ be a function of Y . The **conditional expectation** of $g(Y)$ given $X = x$ is defined as

$$\mathbb{E}[g(Y) | X = x] := \int_{S_Y(x)} g(y) f_{Y|X}(y | x) dy$$

where $x \in S_X$.

- Conditional mean: $\mathbb{E}[Y | X = x] = \int_{S_Y(x)} y \cdot f_{Y|X}(y | x) dy$
- Conditional variance: $\begin{aligned} \text{Var}(Y | X = x) &:= \mathbb{E}[(Y - \mathbb{E}[Y | X = x])^2 | X = x] \\ &= \mathbb{E}[Y^2 | X = x] - (\mathbb{E}[Y | X = x])^2 \end{aligned}$

6 Bivariate Normal Distribution

Definition 6.1 (Bivariate Normal Distribution).

Parameters $\mu_X \in \mathbb{R} \quad \mu_Y \in \mathbb{R} \quad \sigma_X > 0 \quad \sigma_Y > 0 \quad \rho \in [-1, 1]$

X and Y are said to have a **bivariate normal distribution** if the joint pdf is given by

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right]\right)$$

where ρ is the *correlation coefficient* of X and Y .

Theorem 6.1 (Bivariate Normal Distribution).

- *Marginal:* $X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2)$
- *Conditional:* $[X | Y = y] \sim N\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), (1 - \rho^2)\sigma_X^2\right)$
 $[Y | X = x] \sim N\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), (1 - \rho^2)\sigma_Y^2\right)$
- *Covariance:* $\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y$
- *Correlation Coefficient:* $\rho(X, Y) = \rho$

Proof (Covariance of bivariate normal distribution).

Since $X \sim N(\mu_X, \sigma_X^2)$, $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$. Moreover, $\mathbb{E}[Y | X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$. Then,

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}\left[X \cdot \mathbb{E}[Y | X = X]\right] = \rho \frac{\sigma_X}{\sigma_Y} \mathbb{E}[X^2] + \left(\mu_Y - \mu_X \rho \frac{\sigma_X}{\sigma_Y}\right) \mathbb{E}[X] = \rho\sigma_X\sigma_Y + \mu_X\mu_Y \\ \text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \rho\sigma_X\sigma_Y\end{aligned}$$

□

Theorem 6.2.

For a bivariate normal distribution,

$$\text{Independence} \iff \text{Uncorrelation}$$

Proof.

For any bivariate RV, independence implies uncorrelation. Therefore, we only need to prove the converse.

$$\text{Cov}(X, Y) = 0 \implies \rho = 0 \implies [X | Y = y] \sim N(\mu_X, \sigma_X^2) \implies [X | Y = y] = X$$

□